Deepfake Audio Detection: Real vs. Synthetically Cloned Voices

A Generalization-Focused Approach on the ASVspoof 2019 LA Track

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Abstract

Modern text-to-speech (TTS) and voice conversion (VC) systems can generate synthetic speech that is perceptually indistinguishable from genuine human speech, posing a serious threat to telephone banking, voice biometric authentication, and the integrity of public discourse. This project develops an audio deepfake detector trained on the ASVspoof 2019 Logical Access (LA) track, with explicit emphasis on cross-condition generalization rather than raw in-distribution accuracy. We propose to combine the lightweight AASIST-L spectro-temporal graph attention network with on-the-fly RawBoost augmentation, evaluating with the standard Equal Error Rate (EER) and tandem detection cost function (min t-DCF) metrics.

1 Introduction

Audio deepfakes — speech generated by neural TTS or VC systems — have advanced to a quality level where even trained human listeners cannot reliably distinguish them from genuine recordings [1]. The same technology that powers benign products such as voice assistants and audiobook narration can be weaponized for telephone fraud, impersonation of public figures, and bypassing voice-based biometric access control. As neural vocoders such as WaveNet and WaveRNN become widely deployed, the cost of producing a convincing fake utterance approaches zero, while the consequences of a successful spoofing attack scale with the value of the system being attacked.

Most published anti-spoofing countermeasures report sub-1% EER on the ASVspoof 2019 LA evaluation set, suggesting the problem is nearly solved. However, Müller et al. [9] demonstrate that the same models, when evaluated on real-world deepfakes from social media, degrade by 200–1000% in EER — effectively reducing them to random guessing. The conclusion is that the field has, in part, overfit to the studio-quality VCTK-derived ASVspoof distribution. For this project, the goal is therefore not to push the in-distribution EER toward zero, but to build a detector whose architecture, augmentation, and training regime are explicitly designed to generalize beyond the conditions seen during training. Concretely, we will prefer learned raw-waveform front-ends over hand-crafted features (which depend on attack-specific high-frequency artifacts [7]), apply augmentation that simulates realistic channel variability [5], and report multiple random seeds with both EER and min t-DCF.

2 Literature Review

2.1 Benchmarks and Baselines

The ASVspoof 2019 LA track [1] is the de facto benchmark for audio deepfake detection. It is constructed from the VCTK corpus (107 speakers, 16 kHz) and contains 19 spoofing systems spanning

TTS, VC, and hybrid attacks; only 6 attacks (A01–A06) appear in training, while the evaluation set contains 13 mostly-unseen attacks (A07–A19). The associated challenge [2] introduced min t-DCF as the primary metric because it weights each attack by the cost it imposes on a downstream automatic speaker verification system, capturing operational risk that EER alone misses. The official baselines, B01 (CQCC–GMM) and B02 (LFCC–GMM), achieve 9.57% / 8.09% pooled EER and 0.2366 / 0.2116 min t-DCF respectively, while the challenge winner T05 reached 0.22% EER and 0.0069 min t-DCF using a neural network ensemble.

2.2 End-to-End Raw Waveform Architectures

Hand-crafted features such as LFCC and CQCC discard phase information and are tuned to the artifact profile of known vocoders. RawNet2 [3] bypasses this limitation by operating directly on raw waveforms via a fixed sinc-filter front-end, six residual blocks with filter-wise feature-map scaling, and a GRU temporal aggregator. Trained with cross-entropy and Adam (lr $= 10^{-4}$), it achieves 4.66% EER on ASVspoof 2019 LA as a single system and reduces the worst-case A17 (VAE–VC) attack t-DCF by approximately 49% compared to the LFCC baseline. AASIST [4] extends this paradigm by interpreting the RawNet2 encoder output as a 2D feature map and constructing both a spectral and a temporal graph over it. A heterogeneous stacking graph attention layer (HS-GAL) with three distinct edge-type projections, combined with a max-graph operation across parallel branches, achieves 0.83% EER and 0.0275 min t-DCF as a single system. Crucially, the lightweight variant AASIST-L matches near-SOTA performance with only 85K parameters — making it the strongest architecture-per-parameter on this benchmark.

2.3 Self-Supervised Front-Ends and Cross-Condition Generalization

Wang and Yamagishi [7] systematically compare five self-supervised front-ends (wav2vec 2.0 BASE/LARGE, multilingual W2V-Large2, XLS-R, HuBERT-XL [8]) and report that fine-tuned W2V-XLSR + LLGF achieves 0.11% EER on ASVspoof 2019 LA. More importantly, the same model achieves 0.25% EER on ASVspoof 2015 LA, where a comparable LFCC model collapses to 29.42%. A sub-band masking analysis shows that LFCC relies on attack-specific high-frequency cues (5.6–8 kHz), whereas SSL features rely on the linguistically meaningful 0.1–2.4 kHz band, explaining the cross-condition advantage.

2.4 Loss Functions and Reproducibility

Wang and Yamagishi [6] train each configuration of front-end / back-end / loss six times with different seeds and apply Holm–Bonferroni-corrected paired z-tests. They show that single-seed comparisons are statistically unreliable: the same architecture spans 1.92%–3.10% EER across seeds. Their proposed P2SGrad loss — mean-squared error on cosine similarities, with zero hyperparameters — consistently outperforms tuned AM-Softmax and OC-Softmax. The survey by Li et al. [10] corroborates that OC-Softmax and P2SGrad outperform standard cross-entropy on cross-dataset evaluation, and identifies knowledge distillation as the most effective technique for closing the in-domain / out-of-domain gap.

2.5 Data Augmentation

RawBoost [5] is an augmentation method specifically designed for raw-waveform anti-spoofing models. It combines three signal-processing-based perturbations: (i) linear and non-linear convolutive noise via Hammerstein systems and notch filters, (ii) impulsive signal-dependent noise simulating clipping and overflow, and (iii) stationary additive noise. The series combination of (i) + (ii) achieves a 27% relative reduction in min t-DCF and a 44% relative reduction in EER on the ASVspoof 2021 LA telephony evaluation, without requiring any external noise corpus or codec implementation.

2.6 Gap Analysis

Three observations motivate the approach taken in this project. First, the strongest single-system result on ASVspoof 2019 LA (AASIST, 0.83% EER) does not translate to real-world conditions: Müller et al. [9] report 37% EER for the same architecture on in-the-wild deepfakes, indicating that benchmark performance alone is an unreliable proxy for deployment readiness. Second, while fine-tuned wav2vec 2.0 systems achieve the lowest EER on ASVspoof 2019 LA, they require GPU resources (24–40 GB VRAM) that are impractical for a constrained student project, and their min t-DCF can paradoxically be worse than the LFCC baseline despite a lower EER [7]. Third, AASIST-L offers a compelling middle ground: state-of-the-art-competitive accuracy at 85K parameters, an open-source reference implementation, and direct compatibility with on-the-fly RawBoost augmentation. The proposed methodology therefore combines AASIST-L with RawBoost — an architecturally lean, augmentation-rich pipeline whose generalization claims can be validated through multi-seed evaluation and per-attack analysis on the standard A07–A19 protocol.

3 Proposed Methodology

We will train and evaluate a single-system raw-waveform anti-spoofing model based on the AASIST-L architecture with on-the-fly RawBoost augmentation, on the ASVspoof 2019 LA track.

3.1 Dataset and Splits

The ASVspoof 2019 LA dataset [1] will be used unmodified, with the official protocol partitions:

- **Training:** 2,580 bona fide and 22,800 spoofed utterances generated by attacks A01–A06.
- **Development:** 2,548 bona fide and 22,296 spoofed utterances generated by the same six attacks; used for early stopping and hyperparameter selection.
- **Evaluation:** 7,355 bona fide and 63,882 spoofed utterances generated by attacks A07–A19, of which most are unseen during training. This split is the primary measure of generalization to unknown attacks.

All audio will be resampled to 16 kHz mono and randomly cropped or zero-padded to 64,000 samples (approximately 4 seconds) per training example, matching the AASIST input specification.

3.2 Input Representation

The model will operate directly on raw waveform inputs. No hand-crafted spectral features (LFCC, CQCC, MFCC) will be computed at inference time. The first layer of the AASIST-L encoder is a fixed Mel-scaled sinc-convolutional bank of 70 band-pass filters, which produces a learned filterbank-style representation while remaining trainable end-to-end with the rest of the network.

3.3 Model Architecture

The model will follow the AASIST-L specification of Jung et al. [4], layer-by-layer:

1. **Sinc-conv front-end:** 70 fixed Mel-scaled sinc filters, kernel length 129, followed by max-pooling, batch normalization, and SeLU activation.
2. **Residual encoder:** six residual convolutional blocks (first two with 32 filters, last four with 64 filters), each followed by max-pooling and batch normalization. The output is a feature map $F \in R^{C \times S \times T}$ representing channels, spectral bins, and temporal frames.
3. **Graph construction:** two graphs are derived from F. The spectral graph G_s uses temporal-max-pooled node features; the temporal graph G_t uses spectral-max-pooled node features.
4. **Heterogeneous stacking graph attention (HS-GAL):** three distinct projection vectors compute attention weights for intra-spectral, intra-temporal, and cross-domain edges. A learned *stack node*, with uni-directional incoming edges from all other nodes, accumulates cross-domain context analogous to a [CLS] token.
5. **Max graph operation (MGO):** two parallel HS-GAL branches each apply two HS-GAL layers and graph pooling (50% spectral and 30% temporal node removal), with an element-wise maximum taken across branches.
6. **Readout:** the final utterance representation is the concatenation of node-wise max-pooling, node-wise average-pooling, and the stack node embedding.
7. **Output:** a single fully-connected layer projects to 2 logits (bona fide vs. spoof).

The full model contains approximately 85,000 trainable parameters and fits within 8 GB of GPU memory at the chosen batch size.

3.4 Optimizer, Learning Rate, and Loss

The model will be optimized using **Adam** ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with an initial learning rate of 10^{-4}, scheduled by **cosine annealing** over 100 epochs. The loss function will be **weighted cross-entropy**, with a higher weight assigned to the bona fide class to compensate for the approximately 9:1 spoof-to-bona-fide imbalance in the training partition. Batch size will be 24, chosen to fit comfortably within free-tier GPU memory while preserving stable batch statistics.

3.5 Data Augmentation Strategy

RawBoost [5] will be applied on-the-fly inside the training data loader. Each training utterance will pass through the series combination of:

- **Component ①:** Linear and non-linear convolutive noise. A bank of five notch filters with randomly sampled center frequencies $f_c \in [20, 8000]$ Hz and bandwidths $\Delta f \in [100, 1000]$ Hz, followed by a Hammerstein non-linearity that injects controlled harmonic distortion.
- **Component ②:** Impulsive signal-dependent additive noise applied to a randomly chosen subset of $P_{\text{rel}} \in [0, 10]\%$ of samples per utterance.

The component (3) stationary noise of RawBoost will be omitted, since the empirical results in [5] show that it does not improve — and slightly degrades — the series ①+② configuration on LA-style data without ambient noise. Augmentation parameters will be re-sampled per-utterance per-epoch so that the model never sees the same distorted waveform twice.

3.6 Generalization Techniques

The proposed pipeline embeds several generalization-oriented design choices:

- **Fixed sinc filters:** the first-layer center frequencies are not learned, preventing the encoder from overfitting to the limited diversity of the six training attacks [3].
- **Graph pooling as regularization:** 50% spectral and 30% temporal node dropout inside the HS-GAL stacks acts as structured stochastic regularization at the graph level.
- **RawBoost augmentation:** signal-processing-based perturbations expose the model to convolutive and impulsive distortions absent from the clean training set, broadening the input distribution.
- **Multi-seed evaluation:** following [6], the entire training pipeline will be repeated with three different random seeds, and the mean and standard deviation of EER and min t-DCF will be reported.
- **Per-attack reporting:** we will report EER and min t-DCF separately for each of A07–A19, with explicit attention to A17 (VAE–VC), historically the hardest attack [3].

3.7 Evaluation Metrics

Primary evaluation will use the official ASVspoof 2019 LA scoring toolkit. Two metrics will be reported on the evaluation partition:

- **Equal Error Rate (EER, %):** the operating point at which the false acceptance rate and false rejection rate are equal.
- **Minimum tandem Detection Cost Function (min t-DCF):**

  $$\text{t-DCF}^{\text{min}}_{\text{norm}} = \min_{s} \left\{ \beta\, P^{\text{cm}}_{\text{miss}}(s) + P^{\text{cm}}_{\text{fa}}(s) \right\} \tag{1}$$

  where β is set per-attack from the ASV system's false acceptance behaviour, and $P^{\text{cm}}_{\text{miss}}$, $P^{\text{cm}}_{\text{fa}}$ are the miss and false-alarm rates of the countermeasure at threshold s.

Both metrics will be reported for the pooled evaluation set and per-attack.

4 Experimental Progress

5 Preliminary Results

Model	**EER (%)**	**min t-DCF**	**Notes**

Table 1: Preliminary results on the ASVspoof 2019 LA evaluation set. This table will be populated with EER and min t-DCF values once experiments are run; rows will include the official baselines (B01 CQCC–GMM, B02 LFCC–GMM), the proposed AASIST-L baseline (no augmentation), and the proposed AASIST-L + RawBoost configuration, each averaged over three random seeds.

References

[1] X. Wang, J. Yamagishi, M. Todisco, H. Delgado, et al., "ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech," *Computer Speech & Language*, vol. 64, 2020.

[2] M. Todisco, X. Wang, V. Vestman, M. Sahidullah, et al., "ASVspoof 2019: Future horizons in spoofed and fake audio detection," in *Proc. Interspeech*, 2019, pp. 1008–1012.

[3] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, "End-to-end anti-spoofing with RawNet2," in *Proc. ICASSP*, 2021.

[4] J. Jung, H. Heo, H. Tak, H. Shim, J. S. Chung, B. Lee, H. Yu, and N. Evans, "AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks," in *Proc. ICASSP*, 2022.

[5] H. Tak, M. Kamble, J. Patino, M. Todisco, and N. Evans, "RawBoost: A raw data boosting and augmentation method applied to automatic speaker verification anti-spoofing," in *Proc. ICASSP*, 2022.

[6] X. Wang and J. Yamagishi, "A comparative study on recent neural spoofing countermeasures for synthetic speech detection," in *Proc. Interspeech*, 2021.

[7] X. Wang and J. Yamagishi, "Investigating self-supervised front ends for speech spoofing countermeasures," in *Proc. Odyssey*, 2022.

[8] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[9] N. M. Müller, P. Czempin, F. Dieckmann, A. Froghyar, and K. Böttinger, "Does audio deepfake detection generalize?," in *Proc. Interspeech*, 2022.

[10] M. Li, Y. Ahmadiadli, and X.-P. Zhang, "A survey on speech deepfake detection," arXiv:2404.13914, 2024.